

Multiplicity adjustment for Bayesian hypothesis testing via the probability of direction

Margherita Calderan¹, Filippo Gambarota¹, Enrico Toffalini², and Gianmarco Altoè¹

¹Department of Developmental Psychology and Socialization, University of Padova

²Department of General Psychology, University of Padova

Abstract

The probability of direction (pd) has seen rapid adoption in psychological research as an intuitive Bayesian index for quantifying the dominant direction of an effect. In practice, researchers often turn pd into a binary decision rule by applying fixed thresholds (e.g., $pd > 0.975$). When this is done across many parameters, it can increase the risk of misleading conclusions. We argue that this problem stems from neglecting the prior probability of the global null hypothesis, a quantity that becomes increasingly relevant as the number of tested effects grows. To address this, we propose a prior-odds adjustment to pd that explicitly encodes prior beliefs about the prevalence of null effects. This adjustment accounts for multiplicity without modifying the model used in parameter estimation, offers a transparent mechanism for incorporating prior skepticism about the existence of effects, and reduces the risk of false positives in multi-parameter settings. We evaluate its behavior across varying sample sizes and null proportions, and provide an R package to facilitate applied use.

Keywords: probability of direction, multiple testing, prior-odds

Preprint

This manuscript is a **preprint** and has not yet been peer-reviewed. As the content is subject to change, we **welcome** and appreciate any **feedback** you can provide!

Margherita Calderan  <https://orcid.org/0009-0005-5668-5162>

Filippo Gambarota  <https://orcid.org/0000-0002-6666-1747>

Enrico Toffalini  <https://orcid.org/0000-0002-1404-5133>

Gianmarco Altoè  <https://orcid.org/0000-0003-1154-9528>

Code is openly available in the *multibayes* repository on GitHub at <https://github.com/mar-cald/multibayes>. The authors declare that they have no competing interests.

Correspondence concerning this article should be addressed to Margherita Calderan, Department of Developmental Psychology and Socialization, University of Padova, Via Venezia 8, Padova 35131, Italy, Email: margherita.calderan@unipd.it Gianmarco Altoè, Department of Developmental Psychology and Socialization, University of Padova, Via Venezia 8, Padova 35131, Italy, Email: gianmarco.altoe@unipd.it

Over the past two decades, the use of Bayesian methods in psychological research has grown considerably (Jevremov & Pajić, 2024; Pfadt et al., 2025; Van De Schoot et al., 2017). This expansion has, however, been accompanied by practices reminiscent of the “null ritual” in frequentist statistics, where widespread adoption gave way to automatic application rather than principled reasoning (Cohen, 1994; Gigerenzer, 2004). For instance, researchers sometimes treat Bayes Factor thresholds (e.g., $BF > 3$) as dichotomous decision rules, or claim that an effect “exists” because the posterior probability of a directional effect exceeds a conventional benchmark such as 95% (Heck et al., 2023; Hoijtink et al., 2016; Kelter, 2023; Kruschke, 2018; Tendeiro et al., 2024; Tendeiro & Kiers, 2019). While such summaries are convenient, their use as default cutoffs encourages all-or-none interpretations. When these binary judgments are applied repeatedly across many parameters, they generate a multiplicity problem no different in kind from the one long recognized in frequentist statistics.

Yet even there, the problem of multiplicity has been handled inconsistently in practice. Researchers typically account for multiplicity in well-recognized cases, such as post hoc comparisons, but rarely do so when evaluating several effects simultaneously within the same model (Zeevi et al., 2025). This is routine in ANOVA or multiple regression, where main effects, interactions, and multiple predictors each constitute a distinct hypothesis test. Either way, the probability of at least one false positive accumulates with the number of tests performed, regardless of how those tests are framed or grouped (Bender & Lange, 2001; Cramer et al., 2016). Frequentist methods have been developed to address precisely this accumulation, targeting either the Family-Wise Error Rate (FWER; the probability of at least one false positive) or the False Discovery Rate (FDR; the expected proportion of false positives among rejected hypotheses) (Benjamini & Hochberg, 1995; Westfall & Young, 1993). These procedures address whether a spurious result enters the literature, but leave unresolved a subtler consequence of the same selection process. When researchers examine many parameters but highlight only the most extreme estimates, selection systematically inflates reported magnitudes (Altoè et al., 2020; Gelman & Carlin, 2014; Zwet & Cator, 2021).

Bayesian methods can mitigate this overestimation by pulling extreme estimates toward more plausible values, via informative priors (Dawid, 1994; Zwet & Gelman, 2022) and hierarchical modelling (Berry & Hochberg, 1999; Gelman et al., 2012, 2013; Kruschke, 2011). Crucially, however, magnitude bias and false positive accumulation are related but distinct problems. Shrinkage does not, on its own, provide any guarantee about the error properties of a repeated decision rule applied across many parameters: if the same cutoff is applied to many parameters (e.g., declaring an effect whenever the posterior probability exceeds 95%), the probability of spurious findings can remain substantial. Bayesian solutions to this issue typically rely on either simultaneous credible statements (Box & Tiao, 1992; Gelman & Tuerlinckx, 2000; Pruzek, 2016; Schnell et al., 2020) or priors over the hypothesis space that scale with the number of tests (Jeffreys, 1938; Scott & Berger, 2006; Scott & Berger, 2010; Westfall et al., 1997). Despite these available tools, hierarchical modeling alone is still routinely treated as sufficient for multiplicity control, a conflation that leaves the problem only partially addressed (Chen et al., 2020; Erčulj & Štrumbelj, 2015; Hogeveen et al., 2022; Limbachia et al., 2021; Mathar et al., 2022).

This issue becomes particularly salient when posterior summaries such as the probability of direction (pd) are used as decision rules. The pd is a posterior index that quantifies the probability that an effect lies in its dominant direction (Makowski et al., 2019). In recent years, it has increasingly been adopted as a standardized decision criterion, with researchers claiming evidence for an effect when it exceeds fixed thresholds (e.g., $pd > 0.95$ or $pd > 0.975$) (Ben-Artzi et al., 2022; Bramley & Ruggeri, 2022; Brissenden et al., 2023, 2024; Ciranka et al., 2026; Hochman et al., 2025; Makowski et al., 2020; Schurr et al., 2024; Stone-Sabali et al., 2024). A key property of pd is its formal correspondence with the frequentist p -value: for location parameters in exponential family models under a uniform prior, pd equals the complement of the one-sided p -value, and this relationship holds approximately under a broader class of priors (Casella & Berger, 1987; Marsman & Wagenmakers, 2017). Because weakly informative or flat priors are commonly used in applied psychological research (Smid & Winter, 2020; Van De Schoot et al., 2017), thresholding pd

effectively reinstates a decision procedure that is susceptible to multiplicity inflation in much the same way as classical significance testing. Yet this vulnerability has largely gone unrecognized.

To address this gap, we examine the operating characteristics of thresholding rules of the form $pd > c$ under repeated use and consider how the resulting multiplicity problem can be addressed. Specifically, we propose a prior-odds correction for pd grounded in the global null framework of Jeffreys (1938; see also Westfall et al., 1997) and Bayesian variable selection literature (Abraham et al., 2022; Efron et al., 2001; Guo & Heitjan, 2010; Scott & Berger, 2006; Scott & Berger, 2010). This approach requires specifying a prior probability for the global null hypothesis, a context-dependent quantity reflecting the a priori plausibility that all tested effects are simultaneously null. It bears emphasis that the proposed adjustment does not control error rates per se; rather, it makes explicit the prior probability assigned to the global null, allowing the resulting posterior odds to reflect both the observed data and the researcher's substantive beliefs about the prevalence of null effects. When those beliefs are well-calibrated to the data-generating process, error rates are naturally reduced as a consequence.

The paper proceeds as follows. We begin by introducing pd and examining its role as a test for existence, discussing how prior choice shapes its behavior and why more skeptical within-model priors alone cannot resolve the multiplicity problem. We then develop the prior-odds adjustment and evaluate its properties through simulation. A worked example using the accompanying R package (Calderan & Gambarota, 2026) follows, after which we conclude with a general discussion.

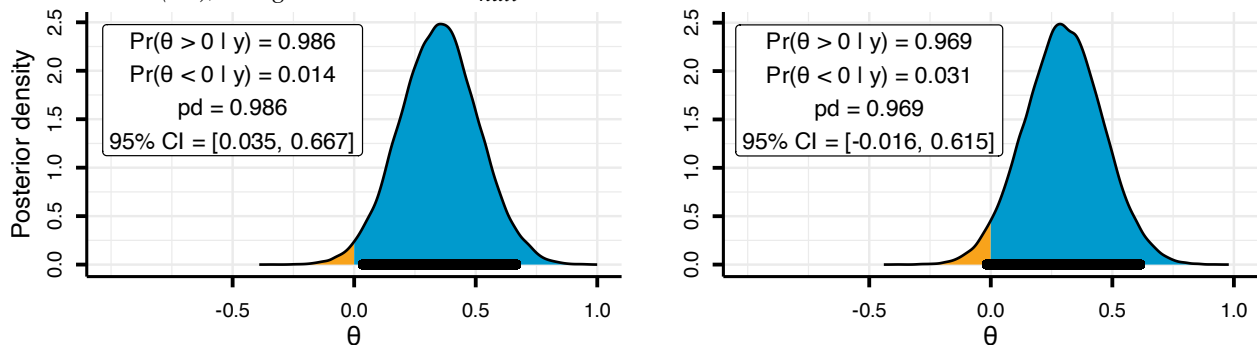
Probability of direction

The probability of direction (pd) is a posterior index that quantifies how strongly the posterior distribution favors one direction of an effect relative to a reference (θ_{null}) value (Makowski et al., 2019). For a continuous posterior, it can be defined as the larger of the two one-sided posterior probabilities: $pd = \max(\Pr(\theta < \theta_{null} | y), \Pr(\theta > \theta_{null} | y))$, and it ranges from 0.50 (no directional preference) to 1 (all on one side). Because pd is derived from the posterior, it depends on the full model specification (likelihood and prior), so the long-run error behavior of pd -based decisions is not a property of pd in isolation but of the overall Bayesian procedure that generates the posterior.

Under flat (or sufficiently diffuse symmetric) priors in typical hypothesis-testing scenarios (e.g., testing means), pd is the posterior probability of the data-favored sign relative to θ_{null} . This value corresponds to the complement of the one-sided p -value (Casella & Berger, 1987; Marsman & Wagenmakers, 2017). In practice, the pd is often used to summarize both the direction and existence of an effect. This interpretation is clearest when phrased as a decision rule $pd > c$: the posterior must assign more than c probability mass to the side of θ_{null} favored by the data.

Figure 1

Posterior distributions illustrating the relationship between the probability of direction (pd) and 95% Credible Interval (CI), using $c = 0.975$ and $\theta_{null} = 0$.



Setting $c = 1 - \alpha/2$ is convenient because it bounds the posterior mass in the opposite tail by $\alpha/2$. For concreteness, suppose the posterior favors a positive effect so that $pd = \Pr(\theta > 0 \mid y)$. An $\alpha = 0.05$ implies $c = 0.975$, therefore the rule $pd > c$ requires that $\Pr(\theta \leq 0 \mid y) < 0.025$, placing less than 2.5% of posterior mass at or below 0. As Figure 1 illustrates, observing $pd > 0.975$ is equivalent to excluding 0 from the central 95% equi-tailed credible interval. When the credible interval excludes the null value, one can infer that the null is relatively implausible (Box & Tiao, 1992).

Given these considerations, although pd and the frequentist p -value target different quantities, it is expected that decision rules of the form $pd > c$ approximate the operating characteristics of familiar frequentist rules such as $p < \alpha$.

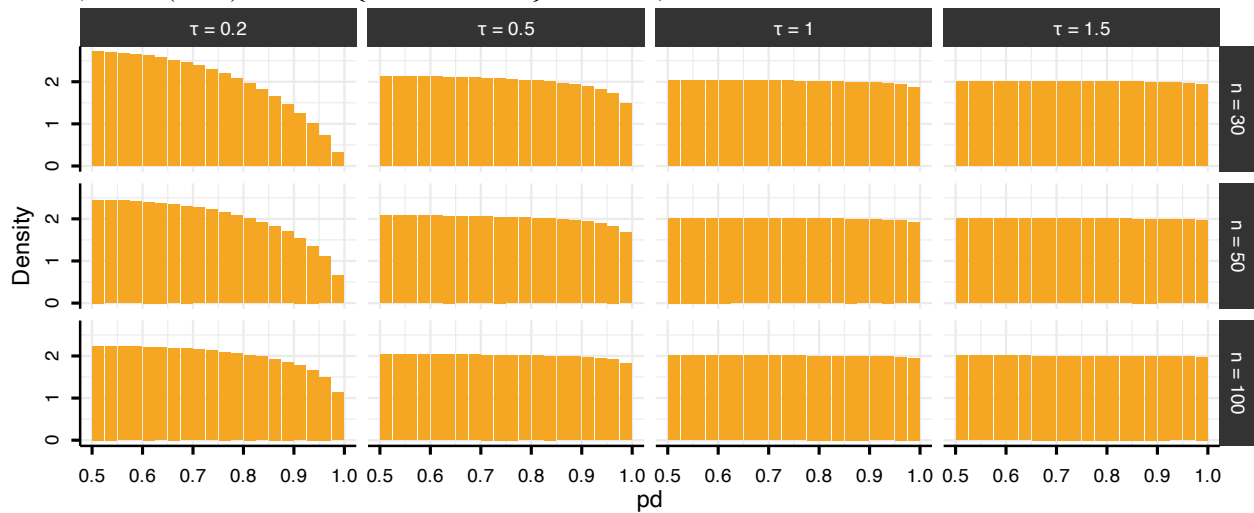
Although we express the decision threshold as $c = 1 - \alpha/2$ (with $\alpha = 0.05$) and set $\theta_{\text{null}} = 0$ throughout for ease of exposition, neither choice is restrictive. The threshold c can be set to any posterior probability cutoff reflecting the researcher's tolerance for uncertainty, and θ_{null} can take any non-zero value, which is particularly relevant when hypotheses involve a minimum effect of practical interest (Lakens et al., 2018).

Multiplicity problem

Under the null hypothesis, p -values are uniformly distributed on $[0, 1]$. Because pd can be mapped to the p -value, this implies a corresponding null reference behavior for pd on $[0.5, 1]$. However, because pd is a posterior probability regarding the sign of θ , its null distribution is not determined by the likelihood alone: it can shift depending on how strongly the prior concentrates probability near 0 (i.e., how strongly the prior “expects” near-null effects). Consequently, even in the absence of multiple testing, assuming the null hypothesis is true, a threshold such $pd > 0.975$ does not correspond to a fixed probability of misleading evidence across priors and sample sizes (Figure 2). For a normal posterior¹, under diffuse priors ($\tau \geq 1$) and/or sufficiently large sample sizes, pd exhibits a distribution with constant density on the interval $[0.5, 1]$. Conversely, the use of a skeptical prior (e.g., $\tau = 0.2$) induces shrinkage toward 0.5, effectively reducing the error rate.

Figure 2

Distribution of the probability of direction (pd) stratified by sample size ($n = 30, 50, 100$; rows) and prior scale ($\theta \sim N(0, \tau^2)$ with $\tau \in \{0.2, 0.5, 1, 1.5\}$; columns). Bin width = 0.025.



¹To illustrate this, we use a conjugate normal-normal model with known variance, in which the posterior distribution and the null distribution of pd admit closed-form expressions. We varied sample size $n \in 30, 50, 100$ and prior scale $\tau \in 0.2, 0.5, 1, 1.5$, ranging from skeptical to relatively diffuse priors.

While skeptical priors reduce the probability that any single test yields an extreme pd , they do not address the issue that, as the number of tests m increases, so does the probability of observing at least one extreme value among $pd_1, \dots, pd_m$. The approach presented below addresses this at the level of the testing procedure rather than the within-model prior, by introducing a separate prior on the global null hypothesis. This allows researchers using uninformative or weakly informative priors to nonetheless incorporate substantive beliefs at the family level.

As we formalize in the next section, a specific condition must be met for pd to be used in this framework. Because pd represents the posterior probability mass on one side of θ_{null} , it maps directly onto the posterior odds between directional hypotheses ($\theta > \theta_{\text{null}}$ vs. $\theta < \theta_{\text{null}}$). These posterior odds equal the Bayes Factor only when the within-model prior is symmetric around θ_{null} , ensuring that each direction is equally plausible a priori, under which the prior odds are unity and the posterior odds reduce to the ratio of marginal likelihoods. This requirement is typically satisfied by the default priors in standard Bayesian software and remains compatible with any degree of prior skepticism regarding effect magnitude.

Prior-Odds adjustment for multiplicity

From posterior probabilities to posterior odds

To connect pd to posterior odds, consider the directional hypotheses $H_+ : \theta > 0$ and $H_- : \theta < 0$. These hypotheses partition the parameter space based on the sign of θ . If the prior for θ is continuous and symmetric about 0, then the two directions are equally plausible a priori: $\Pr(H_+) = \Pr(H_-) = 0.5$.

Without loss of generality, assume the posterior favors H_+ , such that $pd = \Pr(H_+ | y) = \Pr(\theta > 0 | y)$. The complement is then given by $1 - pd = \Pr(\theta \leq 0 | y)$. The posterior odds comparing these two hypotheses are therefore:

$$\frac{pd}{1 - pd} = \frac{\Pr(H_+ | y)}{\Pr(H_{-0} | y)}$$

For a continuous posterior distribution, the probability of any single value (e.g., $\Pr(\theta = 0 | y)$) is 0, which means that one could write $\frac{pd}{1 - pd} = \frac{\Pr(H_+ | y)}{\Pr(H_- | y)}$. However, we retain the notation H_{-0} in the denominator to explicitly indicate that the hypothesis includes the point null value of zero. Applying Bayes' rule, these posterior odds can be decomposed into a Bayes Factor and prior odds:

$$\frac{\Pr(H_+ | y)}{\Pr(H_{-0} | y)} = \text{BF}_{+/-0} \times \frac{\Pr(H_+)}{\Pr(H_{-0})}$$

where $\text{BF}_{+/-0}$ quantifies the relative likelihood of the data under H_+ versus H_{-0} . Under prior symmetry, $\frac{\Pr(H_+)}{\Pr(H_{-0})} = 1$, which yields $\frac{pd}{1 - pd} = \text{BF}_{+/-0}$ (Casella & Berger, 1987; Marsman & Wagenmakers, 2017). Thus, the odds $\frac{pd}{1 - pd}$ computed from posterior draws can be interpreted directly as evidence favoring $\theta > 0$ over $\theta \leq 0$ (or $\theta < 0$ over $\theta \geq 0$).

Multiplicity adjustment via prior-odds

Assessing a single directional claim under a symmetric prior centered at θ_{null} corresponds to prior indifference: $\Pr(H_+) = \Pr(H_{-0}) = 0.5$. When many directional claims are considered jointly, however, treating each with a prior probability of 0.5 implicitly assigns near-certainty to the event that at least one effect lies in its claimed direction. A principled response is to redistribute prior probability toward the null for each individual claim as a function of m (Jeffreys, 1938; Westfall et al., 1997).

Formally, for each test $i = 1, \dots, m$, let $H_{1,i}$ denote the directional claim corresponding to the posterior-dominant sign used to compute $pd_i = \Pr(H_{1,i} | y)$, and let $\Pr(H_{0,i} | y)$ be its complement. The global null hypothesis, $\mathbf{H}_0$, is the event that none of the parameters lie in their favored directions. Under

exchangeability and independence, the prior probability of $\mathbf{H}_0$ can be defined as $\Pi_0 = \pi_0^m$, so specifying Π_0 implies a per-test prior probability

$$\pi_0 = \Pi_0^{1/m}.$$

Thus, π_0 is the marginal prior probability that a given effect is null. Under the same assumptions, the number of null effects follows Binomial(m, π_0), so π_0 corresponds to the expected proportion of nulls. With π_0 in hand, the multiplicity adjustment enters entirely through the prior odds. For each test, the adjusted posterior odds satisfy:

$$\frac{pd_{\text{adj},i}}{1 - pd_{\text{adj},i}} = \frac{pd_i}{1 - pd_i} \times \frac{1 - \pi_0}{\pi_0}$$

which can be expressed as the adjusted posterior probability:

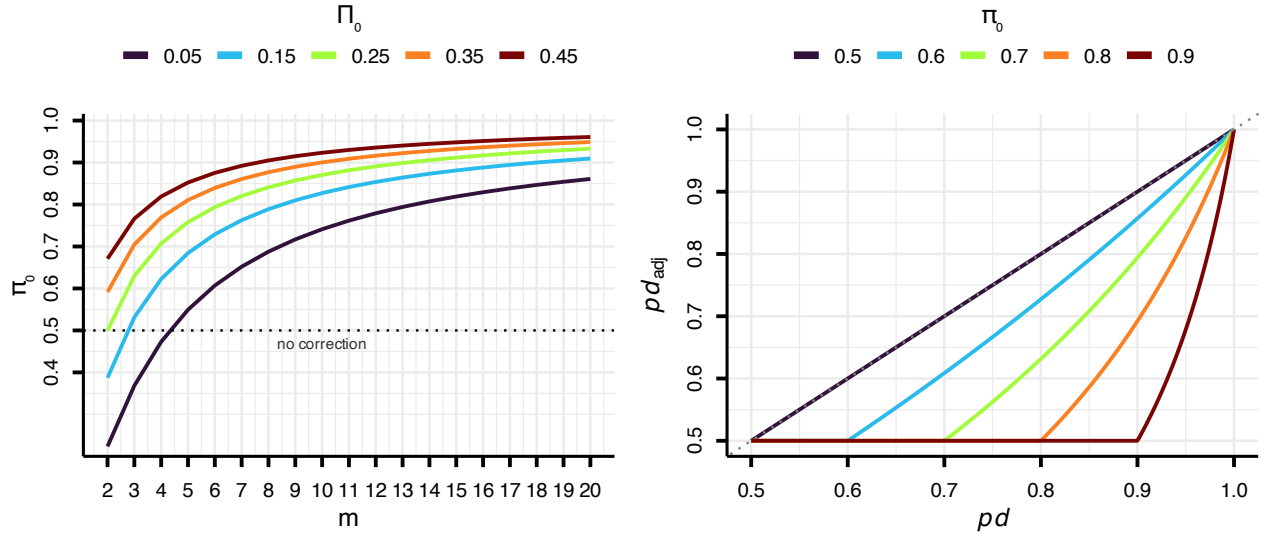
$$pd_{\text{adj},i} = \frac{pd_i \cdot (1 - \pi_0)}{pd_i \cdot (1 - \pi_0) + (1 - pd_i) \cdot \pi_0}.$$

Because the prior is symmetric and pd_i is always the posterior probability of the dominant direction, the factor $(1 - \pi_0)/\pi_0$ acts as a single scalar penalty regardless of the effect's sign.

Researchers encode their expectations through the choice of Π_0 : lower values are appropriate when effects are well established, whereas higher values reflect greater uncertainty, as in exploratory research. For a given Π_0 , larger families lead to higher π_0 ; however, higher pd values are less affected by the correction (see Figure 3).

Figure 3

The left panel shows that the implied prior probability π_0 assigned to each hypothesis $H_{0,i}$ increases with family size m , depending on the specified global null probability Π_0 . The dotted line marks the boundary condition $\pi_0 = 0.5$, below which the prior would favor H_1 a priori. The right panel shows the corresponding prior-odds adjustment of pd for different π_0 values, highlighting the heterogeneous penalty: weak evidence is shrunk much more than strong evidence.



A boundary condition arises when m is small and Π_0 is very low, in which case the formula can yield $\pi_0 < 0.5$, inflating the posterior odds in favor of H_1 rather than penalizing them. We therefore cap π_0 at 0.5, ensuring the adjustment acts exclusively as a conservative safeguard; when no penalty is warranted, $pd_{\text{adj},i} = pd_i$. Furthermore, since $pd \in [0.5, 1]$, the adjusted probability of direction has also bounded below by 0.5.

Simulation Study

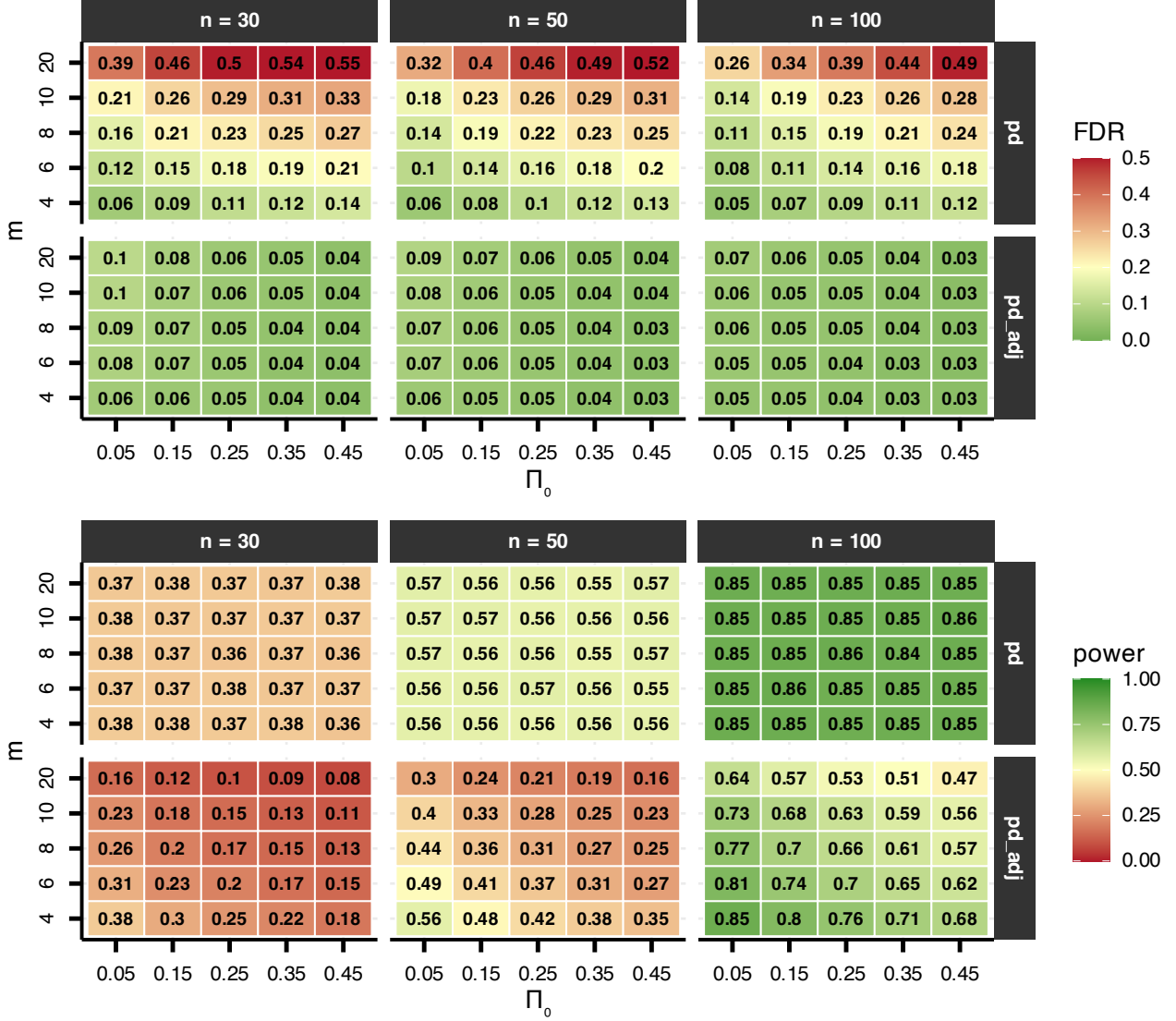
To examine the impact of Π_0 , we conducted a simulation study with a fully crossed factorial design, varying family size $m \in \{4, 6, 8, 10, 20\}$, sample size $n \in \{30, 50, 100, 500\}$, and the researcher-specified global null prior $\Pi_0 \in \{0.05, 0.15, 0.25, 0.35, 0.45\}$, with 10,000 replications per condition. In this first study, Π_0 was set to match the true data-generating null proportion $\pi_{0,true} (= \Pi_0^{1/m})$: at each replication, the number of true null effects was drawn from a Binomial($m, \pi_{0,true}$), and the remaining hypotheses were assigned a non-null effect of $\theta_i = 0.3$. Data were drawn from a standard normal distribution, $y_i \sim \mathcal{N}(\theta_i, 1)$. For every simulated dataset, we computed the posterior distribution of θ using a conjugate normal-normal model with $y_i \mid \theta \sim \mathcal{N}(\theta, 1)$ and prior $\theta \sim \mathcal{N}(0, 2)$.

We considered the alternative hypothesis supported when $pd > 0.975$. Performance was evaluated in terms of the False Discovery Rate (FDR), defined as the proportion of false positives among all hypotheses declared to support the alternative, and average power, defined as the proportion of correctly identified alternative hypotheses (Benjamini & Hochberg, 1995; Izmirlian et al., 2025).

As shown in Figure 4, unadjusted pd -based decisions produced substantial FDR inflation, with error rates increasing as both Π_0 and family size m increased. In contrast, the adjustment procedure performed as intended. In this simulation, $\pi_{0,true} = \pi_0 = \Pi_0^{1/m}$, meaning the specified prior was correctly calibrated; because Π_0 is explicitly incorporated into the correction, FDR did not inflate as a function of m , with values reaching a maximum of 0.10 in small samples. As expected, power decreased as a function of n , Π_0 , and m , reflecting the more stringent penalty imposed under smaller samples, larger families, and higher null probabilities.

Figure 4

False Discovery Rate (FDR) and power of the unadjusted and adjusted pd procedure as a function of the prior probability of the global null (Π_0), the family size m , and the sample size n .



To examine the consequences of prior misspecification, rather than varying Π_0 directly, we parametrized the simulation in terms of the implied per-test null probability $\pi_0 \in \{0.50, 0.55, \dots, 0.95\}$, from which the family-level prior is recovered as $\Pi_0 = \pi_0^m$. The true proportion of null hypotheses $\pi_{0\text{true}} \in \{0.5, 0.6, \dots, 1.0\}$, family size $m \in \{4, 8, 10, 20\}$, and sample size $n \in \{30, 50, 100\}$ were varied independently, with each condition replicated 10,000 times.

As shown in Figure 5, FDR remains near the nominal level across all conditions when m is small ($m = 4$), regardless of sample size, and proves robust even under mild misspecification ($\pi_0 \neq \pi_{0\text{true}}$). A key limitation arises when $\pi_0 = 0.5$ and $m > 4$, for which $\Pi_0 \rightarrow 0$; in small samples, noise can then lead to upward bias in FDR even when $\pi_0 = \pi_{0\text{true}}$. However, larger samples mitigate this issue, with FDR remaining low along the diagonal ($\pi_0 = \pi_{0\text{true}}$). As m increases, FDR rises substantially, particularly when $\pi_{0\text{true}}$ is large and the specified π_0 is small, that is, when the prior assumption underestimates the true proportion of null hypotheses. Consistent with the previous results, the correction reduces average power, with power

decreasing as π_0 increases (Figure 6).

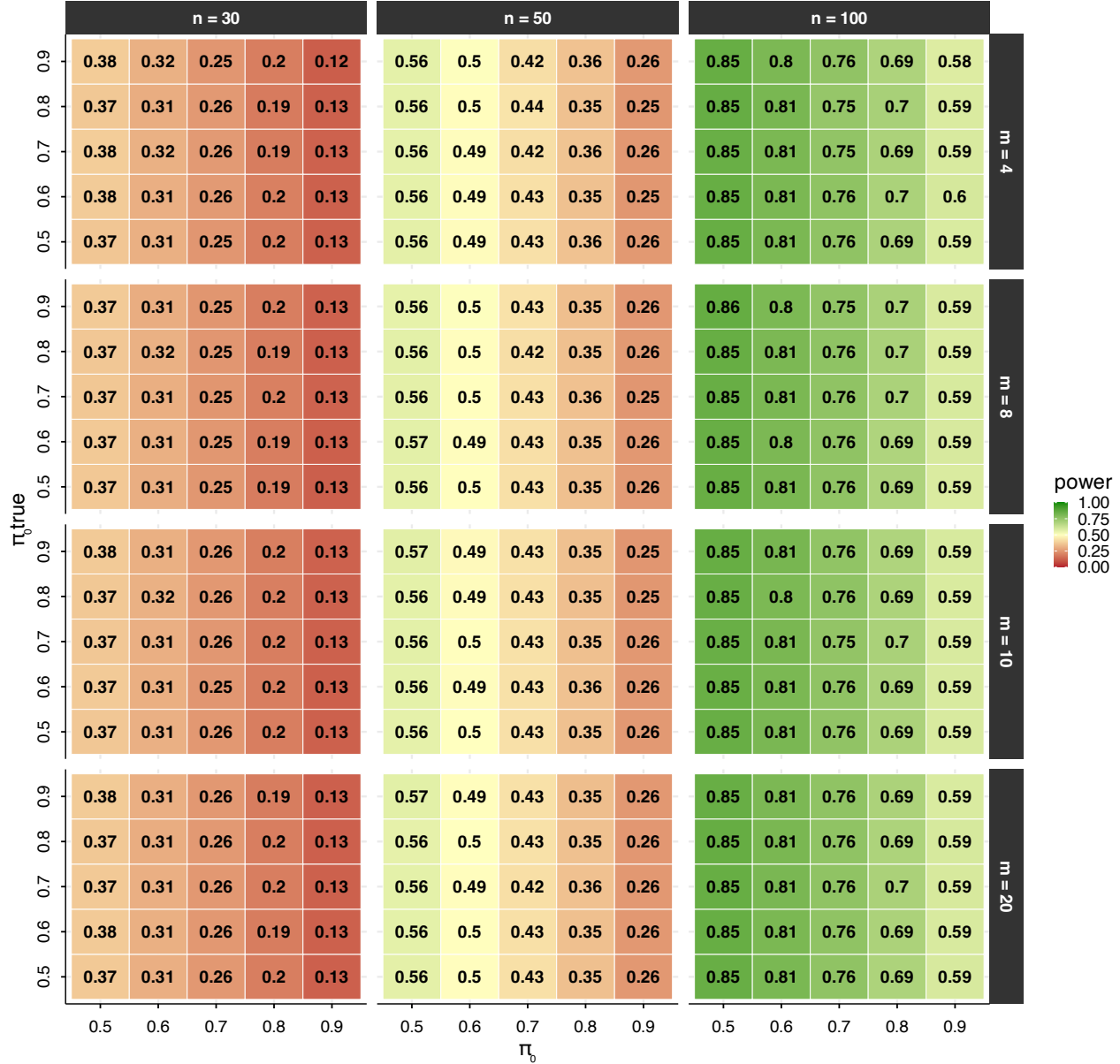
Figure 5

False Discovery Rate (FDR) of the adjusted pd procedure as a function of the prior probability assigned to the null for each test, π_0 , the family size m , and the sample size n . The y-axis represents $\pi_{0,true}$, the true probability that each individual hypothesis is null. Each cell shows the empirical FDR averaged over 10,000 replications.



Figure 6

Average power of the adjusted pd procedure as a function of the prior probability assigned to the null for each test, π_0 , the family size m , and the sample size n . The y-axis represents $\pi_{0,true}$, the true probability that each individual hypothesis is null. Each cell shows the empirical power averaged over 10,000 replications.



Application: worked example with the R Package multibayes

The procedure is implemented in the `multibayes` R package, which provides functions for computing adjusted pd values from either raw posterior draws or a vector of unadjusted pd values. The core function, `pd.adjust()`, accepts posterior draws directly and computes pd values internally when needed. The output includes the adjusted pd values, the tested null value, the implied global null probability, the number of tests m , and the implied per-test null probability. Unadjusted pd values are retained alongside adjusted ones to facilitate direct comparison; when posterior draws are provided as input, the posterior mean for each parameter is also returned.

For this example, we use the openly available dataset from Lin et al. (2020), who investigated the effect of ego depletion on conflict resolution across four studies. In their paradigm, an initial high-demand (vs. low-demand) task was followed by a Stroop task. For simplicity, we focus on Study 1, which tested the effects of congruency and ego depletion (within subjects). The dataset is readily accessible via the `afex` package (Singmann et al., 2016) and was chosen for this reason; this example is not intended as a critique of the original work. This dataset includes data from 253 participants; for this example, we randomly selected 80 participants.

Given the repeated-measures structure of the data and the large number of trials per participant, we fit a multilevel generalized linear model using `brms` (Bürkner, 2017) on reaction time (RT) data. A Lognormal model was specified, with default flat priors; these were adopted not as a principled recommendation, but to illustrate the method's behavior under the most common applied scenario, which also represents the condition where the prior-odds adjustment is most suggested.

We evaluated the effects of congruency (congruent vs. incongruent), previous congruency (congruent vs. incongruent), condition (depletion vs. control), and trial number (scaled within participant and condition). The sequential congruency effect was modeled as the interaction between congruency and previous congruency; the hypothesis that depletion amplifies the congruency effect over time was captured by the three-way congruency $\times$ condition $\times$ trial interaction. This yielded nine hypotheses (Table 1).

Although the theoretical hypotheses are directional (e.g., incongruent trials are expected to slow responses), pd is computed uniformly as $\max(\Pr(\hat{\theta} > \theta_{\text{null}}), \Pr(\hat{\theta} < \theta_{\text{null}}))$ for all hypotheses. This direction-agnostic formulation tests whether the null value falls inside or outside the posterior credible region, without presupposing a predicted direction. Note that hypotheses are tested on the log scale of the posterior. Non-zero null values were specified for all main effects besides condition, and for the sequential congruency effect.

Table 1

Tested hypotheses for RT, evaluated on the log scale of the posterior.

Hypothesis	Null value
H_1 : congruency effect	0.05
H_2 : previous congruency effect	0.05
H_3 : depletion effect	0
H_4 : trial number effect	0.05
H_5 : congruency $\times$ previous congruency interaction	0.05
H_6 : congruency $\times$ condition interaction	0
H_7 : congruency $\times$ trial interaction	0
H_8 : trial $\times$ condition interaction	0
H_9 : congruency $\times$ condition $\times$ trial interaction	0

We retained only correct trials preceded by a correct trial (i.e., error and post-error trials were excluded), yielding 5,965 observations.

```
data_rt <- data_stroop |>
  group_by(id) |>
  filter(acc == 1 & lag(acc) == 1) |>
  ungroup()

# set contrast
```

```

contrasts(data_rt$congruency) <- contr.sum(2)/2
contrasts(data_rt$prev_cong) <- contr.sum(2)/2
contrasts(data_rt$condition) <- contr.sum(2)/2

mod_rt <- brm(formula = rt ~ 1 + congruency * prev_cong +
  congruency * trial * condition +
  (1 + congruency + prev_cong +
    congruency * trial * condition || id),
  family = lognormal(), data = data_rt,
  cores = 3, chains = 3, iter = 8000)

```

Since `pd.adjust()` computes *pd* values automatically from a matrix of posterior draws, we extracted the relevant draws from the model.

```

draws_rt <- as.data.frame(mod_rt)
par_names <- colnames(draws_rt)[2:10]
draws_rt <- draws_rt[, par_names]
pd.adjust(draws = draws_rt, p = 0.02,
  null.value = c(0.05, 0.05, 0.05, 0, 0.05, 0, 0, 0, 0))

```

	mean.est	null.value	pd	pd.adj	p	m	pm
b_congruency1	-0.1440	0.05	1.0000	1.0000	0.02	9	0.6475
b_prev_cong1	-0.0166	0.05	1.0000	1.0000	0.02	9	0.6475
b_trial	0.0119	0.05	1.0000	1.0000	0.02	9	0.6475
b_condition1	0.0064	0.00	0.6509	0.5038	0.02	9	0.6475
b_congruency1:prev_cong1	0.0222	0.05	0.9866	0.9756	0.02	9	0.6475
b_congruency1:trial	-0.0033	0.00	0.7206	0.5840	0.02	9	0.6475
b_congruency1:condition1	-0.0165	0.00	0.9291	0.8770	0.02	9	0.6475
b_trial:condition1	0.0071	0.00	0.8936	0.8205	0.02	9	0.6475
b_congruency1:trial:condition1	-0.0028	0.00	0.5961	0.5000	0.02	9	0.6475
	direction						
b_congruency1	two.sided						
b_prev_cong1	two.sided						
b_trial	two.sided						
b_condition1	two.sided						
b_congruency1:prev_cong1	two.sided						
b_congruency1:trial	two.sided						
b_congruency1:condition1	two.sided						
b_trial:condition1	two.sided						
b_congruency1:trial:condition1	two.sided						

The function was called with $\Pi_0 = p = 0.02$, yielding a per-test prior probability $\pi_0 = pm = p^{1/m} \approx 0.65$. Inspecting the output, three parameters — `b_congruency1`, `b_prev_cong1`, and `b_trial` — show $pd = 1.00$ both before and after adjustment, indicating strong evidence for the alternative hypotheses that is unaffected by the multiplicity correction. The interaction term `b_congruency1:prev_cong1` shows a modest reduction from $pd = 0.987$ to $pd_{\text{adj}} = 0.976$. In contrast, parameters with weaker evidence, such as `b_condition1` ($pd = 0.651$), `b_congruency1:trial` ($pd = 0.721$), and the three-way interaction `b_congruency1:trial:condition1` ($pd = 0.596$), are shrunk substantially toward 0.5 after adjustment.

Discussion

Thresholding pd into a decision rule and applying it repeatedly across many parameters induces a multiplicity problem. Without adjustment, researchers implicitly accept a prior probability for the global null hypothesis (the probability that all tested parameters are simultaneously null) that may be far lower than intended, a concern particularly acute when non-informative or weakly informative priors are used for estimation. The approach developed here addresses this by recasting multiplicity as a problem of prior specification: building on the global null formulation of Jeffreys (1938) and Westfall et al. (1997), the correction requires researchers to explicitly assign a prior probability to the global null event. This makes the procedure a transparent mechanism for linking prior beliefs to decision-making under multiplicity, and underscores the importance of carefully reflecting on the prior probability of the tested hypotheses.

Related corrections have been implemented for specific testing contexts, such as post-hoc tests in JASP (Jong, 2019; van den Bergh et al., 2020). The present procedure extends this logic to pd and provides a general-purpose implementation for applied use, applicable to any posterior-derived pd or directly to posterior draws, including cases where pd values arise from multiple separate analyses within the same manuscript (Blondé et al., 2021; Ciranka et al., 2026). This raises the question of how to define the hypothesis family to which the correction applies (Hochberg & Tamhane, 1987; Shaffer, 1995). Although no universal consensus exists regarding what constitutes a family, we favor the criteria proposed by Westfall et al. (2011), where a family is a set of questions that “*form a natural and coherent unit, [are] considered simultaneously in the decision-making process, and/or present a set over which one plays pick-the-winner*”. Regardless of the chosen definition, authors should clearly specify their chosen family and explain how it serves the study’s substantive goals. We further recommend that researchers justify their choice of prior probability for the global null by drawing on theory, expert elicitation, or domain expertise (O’Hagan, 2019). More generally, a high prior probability suits exploratory settings with many uncertain hypotheses, while a lower value is warranted in confirmatory settings. For applied reporting, the unadjusted pd should accompany one multiplicity-adjusted measure, with the number of tests and the implied global-null prior clearly stated.

Two limitations of the present framework warrant attention. First, the procedure assumes that all hypotheses are exchangeable and independent, assigning each the same prior probability of being null and ignoring dependencies among them. In practice, hypotheses are often dependent, for example when multiple tests rely on overlapping data. Bayesian approaches that explicitly model such dependence or adopt more flexible multiplicity structures have been developed (Gopalan & Berry, 1998; Guo & Heitjan, 2010; Malsiner-Walli & Wagner, 2016; Scott & Berger, 2006; Scott & Berger, 2010; Williams, n.d.), but their implementation requires substantially more modeling effort, limiting their accessibility in applied research.

Second, it does not provide evidence for a point null hypothesis. The procedure evaluates directional claims defined by inequalities (e.g., $\theta > \theta_{\text{null}}$ vs. $\theta \leq \theta_{\text{null}}$), and while the complementary hypothesis includes the null value, that value receives no special status. When the null falls outside the credible interval, one can infer its relative implausibility, but this is fundamentally different from quantifying evidence in its favor (Heck et al., 2023; Ly et al., 2016; Morey & Rouder, 2011). It is worth noting, however, that the prior-odds correction introduced here is not conceptually novel and can be extended to other Bayes Factor approaches. Rather than a method-specific innovation, it reflects a general principle of hypothesis testing: prior probabilities should be combined with all relevant evidence (Dienes, 2016). When joint modeling is feasible, a natural alternative is to integrate evidence across hypotheses via joint posterior probabilities (Box & Tiao, 1992; Gelman & Tuerlinckx, 2000; Pruzek, 2016; Schnell et al., 2020).

As Bayesian methods become increasingly common in psychological research, often in pragmatic workflows relying on default priors and fixed decision thresholds, the implications of multiplicity should be explicitly recognized. Additionally, although priors may be well-calibrated at the level of individual parameters, they do not by themselves account for the prior probability that effects are null when many parameters are considered simultaneously. Ignoring this aspect leads fixed pd thresholds to systematically

overstate evidence for directional effects, especially as the number of tests grows. The adjustment proposed here addresses this gap by incorporating prior beliefs about the prevalence of null effects directly into *pd*-based decisions. By doing so, it provides a principled and transparent way to account for multiplicity without altering the estimation model, yielding more reliable inferences in multi-parameter settings and reducing the risk that apparent effects reflect chance rather than substantive signals.

References

- Abraham, K., Castillo, I., & Roquain, É. (2022). Empirical bayes cumulative l-value multiple testing procedure for sparse sequences. *Electronic Journal of Statistics*, 16(1), 2033–2081.
- Altoè, G., Bertoldo, G., Zandonella Callegger, C., Toffalini, E., Calcagni, A., Finos, L., & Pastore, M. (2020). Enhancing statistical inference in psychological research via prospective and retrospective design analysis. *Frontiers in Psychology*, Volume 10 - 2019. <https://doi.org/10.3389/fpsyg.2019.02893>
- Ben-Artzi, I., Luria, R., & Shahar, N. (2022). Working memory capacity estimates moderate value learning for outcome-irrelevant features. *Scientific Reports*, 12(1), 19677. <https://doi.org/10.1038/s41598-022-21832-x>
- Bender, R., & Lange, S. (2001). Adjusting for multiple testing—when and how? *Journal of Clinical Epidemiology*, 54(4), 343–349. [https://doi.org/10.1016/S0895-4356\(00\)00314-0](https://doi.org/10.1016/S0895-4356(00)00314-0)
- Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1), 289–300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>
- Berry, D. A., & Hochberg, Y. (1999). Bayesian perspectives on multiple comparisons. *Journal of Statistical Planning and Inference*, 82(1), 215–227. [https://doi.org/10.1016/S0378-3758\(99\)00044-0](https://doi.org/10.1016/S0378-3758(99)00044-0)
- Blondé, P., Makowski, D., Sperduti, M., & Piolino, P. (2021). In Medio Stat Virtus: Intermediate levels of mind wandering improve episodic memory encoding in a virtual environment. *Psychological Research*, 85(4), 1613–1625. <https://doi.org/10.1007/s00426-020-01358-5>
- Box, G. E. P., & Tiao, G. C. (1992). *Bayesian inference in statistical analysis* (1st ed.). Wiley. <https://doi.org/10.1002/9781118033197>
- Bramley, N. R., & Ruggeri, A. (2022). Children’s active physical learning is as effective and goal-targeted as adults’. *Developmental Psychology*, 58(12), 2310–2321. <https://doi.org/10.1037/dev0001435>
- Brissenden, J. A., Adkins, T. J., Hsu, Y. T., & Lee, T. G. (2023). Reward influences the allocation but not the availability of resources in visual working memory. *Journal of Experimental Psychology: General*, 152(7), 1825–1839. <https://doi.org/10.1037/xge0001370>
- Brissenden, J. A., Scerbak, T., Albin, R. L., & Lee, T. G. (2024). Motivational vigor in parkinson’s disease requires the short and long duration response to levodopa. *Movement Disorders*, 39(1), 76–84. <https://doi.org/10.1002/mds.29659>
- Bürkner, P.-C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software*, 80(1), 1–28. <https://doi.org/10.18637/jss.v080.i01>
- Calderan, M., & Gambarota, F. (2026). *Multibayes: Bayesian multiplicity adjustments*. <https://github.com/mar-cald/multibayes>
- Casella, G., & Berger, R. L. (1987). Reconciling bayesian and frequentist evidence in the one-sided testing problem. *Journal of the American Statistical Association*, 82(397), 106–111. <https://doi.org/10.1080/01621459.1987.10478396>
- Chen, G., Taylor, P. A., Cox, R. W., & Pessoa, L. (2020). Fighting or embracing multiplicity in neuroimaging? Neighborhood leverage versus global calibration. *NeuroImage*, 206, 116320. <https://doi.org/10.1016/j.neuroimage.2019.116320>
- Ciranka, S., Chevallier, C., & Nettle, D. (2026). Low socioeconomic status amplifies the perceived rarity of large rewards. *Cognition*, 270, 106409. <https://doi.org/10.1016/j.cognition.2025.106409>

- Cohen, J. (1994). The earth is round ($p < .05$). *American Psychologist*, 49(12), 997–1003. <https://doi.org/10.1037/0003-066X.49.12.997>
- Cramer, A. O. J., van Ravenzwaaij, D., Matzke, D., Steingroever, H., Wetzels, R., Grasman, R. P. P. P., Waldorp, L. J., & Wagenmakers, E.-J. (2016). Hidden multiplicity in exploratory multiway ANOVA: Prevalence and remedies. *Psychonomic Bulletin & Review*, 23(2), 640–647. <https://doi.org/10.3758/s13423-015-0913-5>
- Dawid, A. P. (1994). Selection paradoxes of bayesian inference. *Lecture Notes-Monograph Series*, 24, 211–220.
- Dienes, Z. (2016). How bayes factors change scientific practice. *Journal of Mathematical Psychology*, 72, 78–89. <https://doi.org/https://doi.org/10.1016/j.jmp.2015.10.003>
- Efron, B., Tibshirani, R., Storey, J. D., & Tusher, V. (2001). Empirical bayes analysis of a microarray experiment. *Journal of the American Statistical Association*, 96(456), 1151–1160. <https://doi.org/10.1198/016214501753382129>
- Erčulj, F., & Štrumbelj, E. (2015). Basketball shot types and shot success in different levels of competitive basketball. *PLOS ONE*, 10(6), 1–14. <https://doi.org/10.1371/journal.pone.0128885>
- Gelman, A., & Carlin, J. (2014). Beyond power calculations: Assessing type s (sign) and type m (magnitude) errors. *Perspectives on Psychological Science*, 9(6), 641–651. <https://doi.org/10.1177/1745691614551642>
- Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B. (2013). *Bayesian data analysis* (3rd ed.). Chapman; Hall/CRC. <https://doi.org/10.1201/b16018>
- Gelman, A., Hill, J., & Yajima, M. (2012). Why we (usually) don't have to worry about multiple comparisons. *Journal of Research on Educational Effectiveness*, 5(2), 189–211. <https://doi.org/10.1080/19345747.2011.618213>
- Gelman, A., & Tuerlinckx, F. (2000). Type s error rates for classical and bayesian single and multiple comparison procedures. *Computational Statistics*, 15(3), 373–390. <https://doi.org/10.1007/s001800000040>
- Gigerenzer, G. (2004). Mindless statistics. *The Journal of Socio-Economics*, 33(5), 587–606. <https://doi.org/https://doi.org/10.1016/j.socec.2004.09.033>
- Gopalan, R., & Berry, D. A. (1998). Bayesian multiple comparisons using dirichlet process priors. *Journal of the American Statistical Association*, 93(443), 1130–1139. <http://www.jstor.org/stable/2669856>
- Guo, M., & Heitjan, D. F. (2010). Multiplicity-calibrated Bayesian hypothesis tests. *Biostatistics*, 11(3), 473–483. <https://doi.org/10.1093/biostatistics/kxq012>
- Heck, D. W., Boehm, U., Böing-Messing, F., Bürkner, P.-C., Derks, K., Dienes, Z., Fu, Q., Gu, X., Karimova, D., Kiers, H. A. L., Klugkist, I., Kuiper, R. M., Lee, M. D., Leenders, R., Leplaa, H. J., Linde, M., Ly, A., Meijerink-Bosman, M., Moerbeek, M., ... Hoijtink, H. (2023). A review of applications of the Bayes factor in psychological research. *Psychological Methods*, 28(3), 558–579. <https://doi.org/10.1037/met0000454>
- Hochberg, Y., & Tamhane, A. C. (1987). *Multiple Comparison Procedures* (1st ed.). Wiley. <https://doi.org/10.1002/9780470316672>
- Hochman, S., Havedanloo, R., Heysieattalab, S., & Soltanlou, M. (2025). How does language modulate the association between number and space? A registered report of a cross-cultural study of the spatial-numerical association of response codes effect. *Journal of Experimental Psychology: General*, 154(2), 305–324. <https://doi.org/10.1037/xge0001653>
- Hogeveen, J., Mullins, T. S., Romero, J. D., Eversole, E., Rogge-Obando, K., Mayer, A. R., & Costa, V. D. (2022). The neurocomputational bases of explore-exploit decision-making. *Neuron*, 110(11), 1869–1879.e5. <https://doi.org/10.1016/j.neuron.2022.03.014>
- Hoijtink, H., Kooten, P. van, & Hulsker, K. (2016). Why bayesian psychologists should change the way they use the bayes factor. *Multivariate Behavioral Research*, 51(1), 2–10.
- Izmirlian, G., Sirota, L. A., Berger, V. W., & Kipnis, V. (2025). The fundamentals of multiplicity adjustment

- in biostatistics. *JNCI Monographs*, 2025(68), 10–13. <https://doi.org/10.1093/jncimonographs/lgae050>
- Jeffreys, H. (1938). Significance tests when several degrees of freedom arise simultaneously. *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences*, 165(921), 161–198. <https://doi.org/10.1098/rspa.1938.0052>
- Jevremov, T., & Pajić, D. (2024). Bayesian method in psychology: A bibliometric analysis. *Current Psychology*, 43(10), 8644–8654.
- Jong, T. de. (2019). A bayesian approach to the correction for multiplicity. PsyArXiv preprint. <https://doi.org/10.31234/osf.io/s56mk>
- Kelter, R. (2023). How to choose between different bayesian posterior indices for hypothesis testing in practice. *Multivariate Behavioral Research*, 58(1), 160–188.
- Kruschke, J. K. (2011). Bayesian assessment of null values via parameter estimation and model comparison. *Perspectives on Psychological Science*, 6(3), 299–312. <https://doi.org/10.1177/1745691611406925>
- Kruschke, J. K. (2018). Rejecting or Accepting Parameter Values in Bayesian Estimation. *Advances in Methods and Practices in Psychological Science*, 1(2), 270–280. <https://doi.org/10.1177/2515245918771304>
- Lakens, D., Scheel, A. M., & Isager, P. M. (2018). Equivalence testing for psychological research: A tutorial. *Advances in Methods and Practices in Psychological Science*, 1(2), 259–269. <https://doi.org/10.1177/2515245918770963>
- Limbachia, C., Morrow, K., Khibovska, A., Meyer, C., Padmala, S., & Pessoa, L. (2021). Controllability over stressor decreases responses in key threat-related brain areas. *Communications Biology*, 4(1), 42. <https://doi.org/10.1038/s42003-020-01537-5>
- Lin, H., Saunders, B., Friese, M., Evans, N. J., & Inzlicht, M. (2020). Strong effort manipulations reduce response caution: A preregistered reinvention of the ego-depletion paradigm. *Psychological Science*, 31(5), 531–547. <https://doi.org/10.1177/0956797620904990>
- Ly, A., Verhagen, J., & Wagenmakers, E.-J. (2016). Harold jeffreys's default bayes factor hypothesis tests: Explanation, extension, and application in psychology. *Journal of Mathematical Psychology*, 72, 19–32. <https://doi.org/https://doi.org/10.1016/j.jmp.2015.06.004>
- Makowski, D., Ben-Shachar, M. S., Chen, S. H. A., & Lüdtke, D. (2019). Indices of effect existence and significance in the bayesian framework. *Frontiers in Psychology, Volume 10 - 2019*. <https://doi.org/10.3389/fpsyg.2019.02767>
- Makowski, D., Sperduti, M., Blondé, P., Nicolas, S., & Piolino, P. (2020). The heart of cognitive control: Cardiac phase modulates processing speed and inhibition. *Psychophysiology*, 57(3), e13490. <https://doi.org/https://doi.org/10.1111/psyp.13490>
- Malsiner-Walli, G., & Wagner, H. (2016). Comparing spike and slab priors for bayesian variable selection. *Austrian Journal of Statistics*, 40(4). <https://doi.org/10.17713/ajs.v40i4.215>
- Marsman, M., & Wagenmakers, E.-J. (2017). Three insights from a bayesian interpretation of the one-sided p value. *Educational and Psychological Measurement*, 77(3), 529–539. <https://doi.org/10.1177/0013164416669201>
- Mathar, D., Erfanian Abdoust, M., Marrenbach, T., Tuzsus, D., & Peters, J. (2022). The catecholamine precursor tyrosine reduces autonomic arousal and decreases decision thresholds in reinforcement learning and temporal discounting. *PLOS Computational Biology*, 18(12). <https://doi.org/10.1371/journal.pcbi.1010785>
- Morey, R. D., & Rouder, J. N. (2011). Bayes factor approaches for testing interval null hypotheses. *Psychological Methods*, 16(4), 406–419. <https://doi.org/10.1037/a0024377>
- O'Hagan, A. (2019). Expert knowledge elicitation: Subjective but scientific. *The American Statistician*, 73(sup1), 69–81. <https://doi.org/10.1080/00031305.2018.1518265>
- Pfadt, J. M., Bartoš, F., Godmann, H. R., Waaijers, M., Groot, L., Heo, I., et al. (2025, October 30). A methodological metamorphosis: The rapid rise of bayesian inference and open science practices in psychology. OSF Preprints. https://doi.org/10.31234/osf.io/ck3js_v1

- Pruzek, R. M. (2016). *An introduction to bayesian inference and its applications* (pp. 305–334). Routledge.
- Schnell, P. M., Fiecas, M., & Carlin, B. P. (2020). **credsubs** : Multiplicity-adjusted subset identification. *Journal of Statistical Software*, 94(7). <https://doi.org/10.18637/jss.v094.i07>
- Schurr, R., Reznik, D., Hillman, H., Bhui, R., & Gershman, S. J. (2024). Dynamic computational phenotyping of human cognition. *Nature Human Behaviour*, 8(5), 917–931. <https://doi.org/10.1038/s41562-024-01814-x>
- Scott, J. G., & Berger, J. O. (2006). An exploration of aspects of bayesian multiple testing. *Journal of Statistical Planning and Inference*, 136(7), 2144–2162.
- Scott, J. G., & Berger, J. O. (2010). Bayes and empirical-Bayes multiplicity adjustment in the variable-selection problem. *The Annals of Statistics*, 38(5), 2587–2619. <https://doi.org/10.1214/10-AOS792>
- Shaffer, J. P. (1995). Multiple hypothesis testing. *Annual Review of Psychology*, 46, 561–584. <https://doi.org/10.1146/annurev.ps.46.020195.003021>
- Singmann, H., Bolker, B., Westfall, J., & Aust, F. (2016). *Afex: Analysis of factorial experiments*. <https://CRAN.R-project.org/package=afex>
- Smid, S. C., & Winter, S. D. (2020). Dangers of the defaults: A tutorial on the impact of default priors when using bayesian SEM with small samples. *Frontiers in Psychology, Volume 11 - 2020*. <https://doi.org/10.3389/fpsyg.2020.611963>
- Stone-Sabali, S., Uanhoro, J. O., McClain, S., Bernard, D., Makari, S., & Chapman-Hilliard, C. (2024). Impostorism and Psychological Distress among College Students of Color: A moderation analysis of Shame-proneness, Race, Gender, and Race-Gender Interactions. *Current Psychology*, 43(4), 3632–3648. <https://doi.org/10.1007/s12144-023-04579-0>
- Tendeiro, J. N., & Kiers, H. A. (2019). A review of issues about null hypothesis bayesian testing. *Psychological Methods*, 24(6), 774.
- Tendeiro, J. N., Kiers, H. A., Hoekstra, R., Wong, T. K., & Morey, R. D. (2024). Diagnosing the misuse of the bayes factor in applied research. *Advances in Methods and Practices in Psychological Science*, 7(1), 25152459231213371.
- Van De Schoot, R., Winter, S. D., Ryan, O., Zondervan-Zwijnenburg, M., & Depaoli, S. (2017). A systematic review of bayesian articles in psychology: The last 25 years. *Psychological Methods*, 22(2), 217.
- van den Bergh, D., van Doorn, J., Marsman, M., Draws, T., van Kesteren, E.-J., Derks, K., Dablander, F., Gronau, Q. F., Kucharský, Š., Gupta, A. R. K. N., Sarafoglou, A., Voelkel, J. G., Stefan, A., Ly, A., Hinne, M., Matzke, D., & Wagenmakers, E.-J. (2020). A tutorial on conducting and interpreting a Bayesian ANOVA in JASP. *L'Année Psychologique*, 120(1), 73–96. <https://doi.org/10.3917/anpsy1.201.0073>
- Westfall, P. H., Johnson, W. O., & Utts, J. M. (1997). A bayesian perspective on the bonferroni adjustment. *Biometrika*, 84(2), 419–427.
- Westfall, P. H., Tobias, R. D., & Wolfinger, R. D. (2011). *Multiple comparisons and multiple tests using SAS, second edition*. SAS Institute.
- Westfall, P. H., & Young, S. S. (1993). *Resampling-based multiple testing: Examples and methods for p-value adjustment*. Wiley.
- Williams, D. R. (n.d.). *Many mixture components, oh my: Extending the spike and slab to bayesian hypothesis testing with multinoulli indicators*. <https://doi.org/10.31234/osf.io/xcvby>
- Zeevi, Y., Catzman, S., Benjamini, Y., & Mudrik, L. (2025). Correction for multiple comparisons should be ubiquitous. *Nature Human Behaviour*, 9(12), 2407–2408. <https://doi.org/10.1038/s41562-025-02313-3>
- Zwet, E. van, & Cator, E. A. (2021). The significance filter, the winner's curse and the need to shrink. *Statistica Neerlandica*, 75(4), 437–452. <https://doi.org/10.1111/stan.12241>
- Zwet, E. van, & Gelman, A. (2022). A proposal for informative default priors scaled by the standard error of estimates. *The American Statistician*, 76(1), 1–9. <https://doi.org/10.1080/00031305.2021.1938225>